SPIN COHERENCE TIME OPTIMIZATION USING A JACOBIAN-BASED MAPPING METHOD

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Abstract

A sufficiently long Spin Coherence Time (SCT) is a key requirement for precision storage-ring measurements of the charged-particle electric dipole moment (EDM). In this work, we formulate a practical SCT optimization procedure based on a Jacobian (mapping) approach. COSY-INFINITY simulations are used to build linear response models for (i) chromatic observables (ξ_x, ξ_y, η_1) and (ii) spin-tune deviations for a set of phase-space probes. The extracted response matrices are then combined into a direct spin–chromaticity transfer operator, enabling a plane-by-plane tuning strategy that enforces cross-plane constraints. The method is applied to several magnetic and electrostatic lattice variants and is used to quantify the impact of fringe fields and non-linearities via a two-amplitude probe scheme.

INTRODUCTION

Spin coherence time is limited by the spread of spin precession frequencies across the stored beam phase space. In realistic rings, the dominant contributions arise from chromatic and geometric effects driven by emittance and momentum spread. Sextupole families are the standard tool to control second-order dynamics; however, tuning for SCT while simultaneously enforcing optical requirements (e.g., prescribed ξ_x, ξ_y, and longitudinal integral η_1) is a multi-objective problem.

Here we use a Jacobian-based mapping method that turns this problem into a sequence of linear solves based on response matrices extracted from tracking data.

JACOBIAN-BASED MAPPING METHOD

Machine state and observables

For each lattice configuration, COSY-INFINITY tracking data are collected for a scan of sextupole family settings. Each machine state is labeled by a vector of sextupole control currents $\vec{I}$. We consider:

- chromatic observables $\vec{\xi} = (\xi_x, \xi_y, \eta_1)^T$;
- spin-tune deviations for phase-space probes $\Delta\vec{\nu}_s = (\Delta\nu_x, \Delta\nu_y, \Delta\nu_z)^T$.

The probe scheme uses two amplitude sets ("P0" nominal and "P1" increased offsets) to diagnose non-linear behavior.

Linear response model

Around the scanned range, we approximate both the chromatic observables and the spin response by a linear model:

$$\vec{\xi} = \mathbf{R}_{int}\vec{I} + \vec{\xi}_{nat}, \qquad \Delta\vec{\nu}_s = \mathbf{R}_{spin}\vec{I} + \Delta\vec{\nu}_{s,nat}, \tag{1}$$

where $\vec{\xi}_{nat}$ and $\Delta\vec{\nu}_{s,nat}$ are the natural (sextupoles-off) intercepts. The matrices $\mathbf{R}_{int}$ and $\mathbf{R}_{spin}$ are extracted using ordinary least squares over the scan dataset.

Control law and transfer operator

For a target chromatic state $\vec{\xi}_t$, the required sextupole setting is computed as

$$\vec{I} = \mathbf{R}_{int}^{-1}\left(\vec{\xi}_t - \vec{\xi}_{nat}\right), \tag{2}$$

and substituted into Eq. (1) to obtain a direct map from chromaticity to spin response:

$$\Delta\vec{\nu}_s = \underbrace{\left(\mathbf{R}_{spin}\mathbf{R}_{int}^{-1}\right)}_{\mathbf{S}}\left(\vec{\xi}_t - \vec{\xi}_{nat}\right) + \Delta\vec{\nu}_{s,nat}. \tag{3}$$

The operator $\mathbf{S}$ quantifies spin-tune sensitivity to controlled chromatic changes and is convenient for comparing different lattice models and field representations.

Decoupled plane-by-plane scans

To isolate the dominant mechanisms, we perform scans in ξ_x, ξ_y, and η_1 while constraining the other two components to zero:

$$\begin{aligned}(\xi_x, \xi_y, \eta_1) &= (\xi_x, 0, 0),\\ (\xi_x, \xi_y, \eta_1) &= (0, \xi_y, 0),\\ (\xi_x, \xi_y, \eta_1) &= (0, 0, \eta_1).\end{aligned} \tag{4}$$

In practice, this means solving Eq. (2) for each scan point, with $\vec{\xi}_t$ constructed according to Eq. (4).

KEY RESULT: SPIN–CHROMATICITY TRANSFER CURVES

The central output of the mapping workflow is the predicted spin-tune response $\Delta\nu_s$ versus target chromatic settings, computed from Eqs. (2)–(3). For each scan axis, we enforce the cross-plane constraints (Eq. 4) and compare the "No fringe" and "With fringe" field models for two probe amplitudes (P0 and P1). Representative results for a magnetic lattice ("magnetic_2p") are shown in Fig. 1–3.

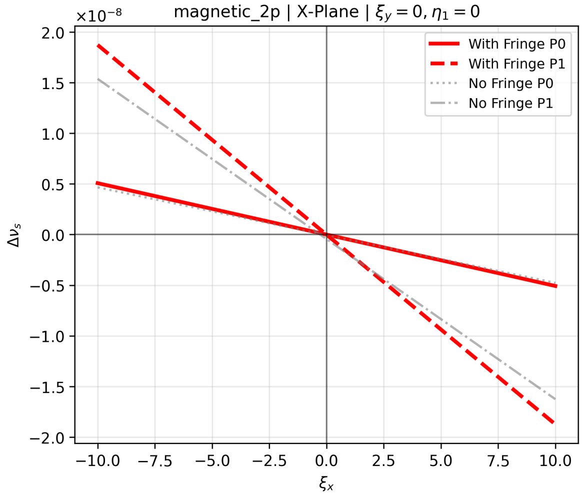

Figure 1: "magnetic_2p": $\Delta\nu_s$ vs ξ_x at $\xi_y = 0,\ \eta_1 = 0$.

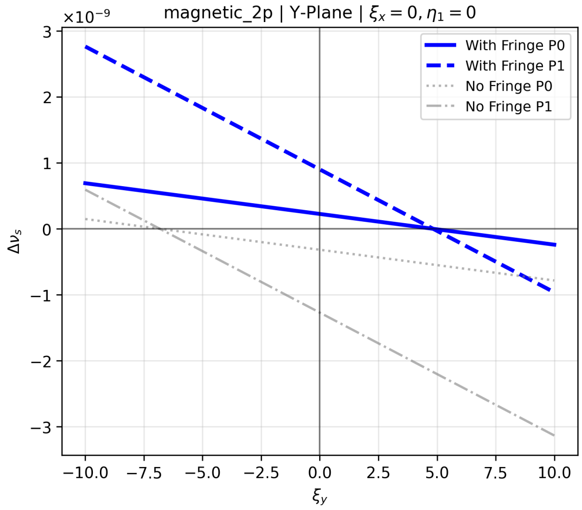

Figure 2: "magnetic_2p": $\Delta\nu_s$ vs ξ_y at $\xi_x = 0,\ \eta_1 = 0$.

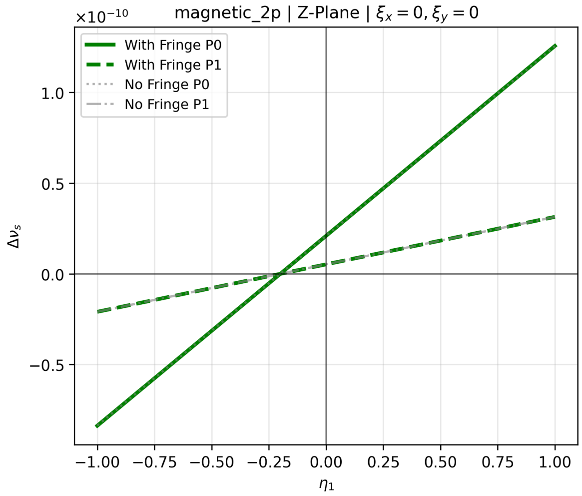

Figure 3: "magnetic_2p": $\Delta\nu_s$ vs η_1 at $\xi_x = 0,\ \xi_y = 0$.

RESULTS AND DISCUSSION

For each lattice, response matrices $\mathbf{R}_{int}$ and $\mathbf{R}_{spin}$ are built independently for two field representations ("No fringe" and "With fringe") and for both probe amplitudes (P0 and P1). The comparison of resulting $\mathbf{S}$ operators provides:

- a quantitative measure of how realistic fringe fields distort the effective spin–chromaticity transfer;
- a non-linearity diagnostic through the difference between P0 and P1 responses;
- a practical tuning workflow, where sextupole settings are computed from Eq. (2) while enforcing constraints in Eq. (4).

A self-consistency check is performed by reconstructing sextupole settings from measured chromatic observables via Eq. (2) and then predicting the spin response using Eq. (1), verifying that higher-order terms are small over the studied range.

CONCLUSION

A Jacobian-based mapping method for SCT-oriented sextupole tuning has been formulated and applied to a set of COSY-INFINITY lattice models. The approach provides an explicit spin–chromaticity transfer operator $\mathbf{S}$ and supports constrained scans in ξ_x, ξ_y, and η_1. The framework naturally incorporates comparisons between fringe/no-fringe field models and between nominal and large-amplitude probes, enabling a structured assessment of robustness and non-linear effects in SCT optimization.

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